Name: _____

TOPIC TEST 1

Chapter 10: Properties of derivatives

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 6 multiple-choice questions (6 marks)
- Part B: 10 short answer questions (34 marks)
- Part C: 6 analysis questions (25 marks)

Part A

6 multiple-choice questions
1 mark each: 6 marks
Circle the correct answer.

1 When a function is increasing, the derivative is:

A decreasing

B negative

C positive

D 0

E increasing

[1 mark]

2 $y = x^3 + 2x^2 - 6x + 7$. At $x = -2$, y is:

A increasing

B stationary

C negative

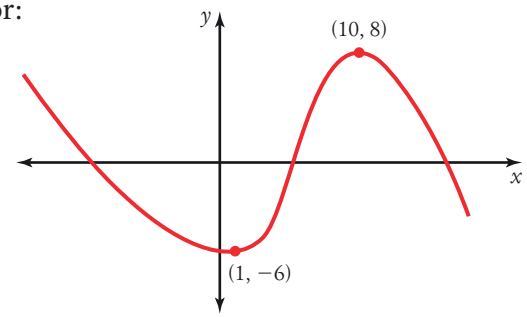
D decreasing

E 0

[1 mark]

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3 For the function shown on the right, the derivative is negative for:



A $1 < x < 10$

B $x > 10$ only

C $x < 1$ and $x > 10$

D $-6 < x < 8$

E $x < 1$ only

[1 mark]

4 If $f(x) = 4x^2 - 3x + 7$, then $f'(3) =$

A 34

B 21

C 28

D 27

E 24

[1 mark]

5 The rate of change for $f(x) = 5x^2 - 12x + 15$ at $x = 2$ is:

A 32

B 9

C 15

D 23

E 8

[1 mark]

6 The antiderivative of $20x^4$ is:

A $4x^5 + c$

B $5x^3 + c$

C $20x^5 + c$

D $5x^5 + c$

E $80x^3 + c$

[1 mark]

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Part B

10 short answer questions
34 marks
Show your working where appropriate.

1 State whether the function $f(x) = 2x^3 + 17x^2 - 12x + 8$ is increasing, decreasing or stationary at:

a $x = 1$ _____

b $x = -2$ _____

c $x = \frac{1}{3}$ _____

[7 marks]

2 $p(x) = 2x^4 - 7x^2 + 8x - 9$. Find:

a $p'(x)$ _____

b $p'(-3)$ _____

[2 marks]

3 If $g(x) = -4x^3 + 2x^2 + 5x - 3$, calculate:

a $5g'(x)$

b $\frac{2}{3}g'(x)$

[3 marks]

4 If $f'(x) = 4x^3 - 3x^2 - x$ and $g'(x) = -3x^2 + 4x + 5$, calculate the derivative of $2f(x) - 7g(x)$.

[4 marks]

5 Find the derivatives of the following functions.

a $p(x) = 7x^5 - 8x^3 + 5x + 11$

b $g(u) = 14 + 5u^3 - 6u - 4u^2$

[2 marks]

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6 Find the derivative of the function $y = (2x^3 + 5x)^2$.

[2 marks]

7 Find the derivative of the function $y = (3x - 2)^3$.

[2 marks]

8 $f(x) = 4x^2 - 3x + 5$.

a Find the average rate of change of from $x = 1$ to $x = 5$.

b Find the instantaneous rate of change at:

i $x = 1$ _____

ii $x = 5$ _____

iii $x = 3$ _____

c Compare the average rate of change and the instantaneous rates of change.

[8 marks]

9 Find two antiderivatives of $f(x) = 8x^3 - 4x + 7$.

[2 marks]

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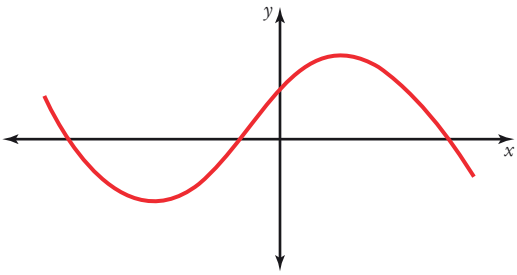
10 Find y if $\frac{dx}{dy} = 4 - 6x^2 - 2x^5$

[2 marks]

Part C

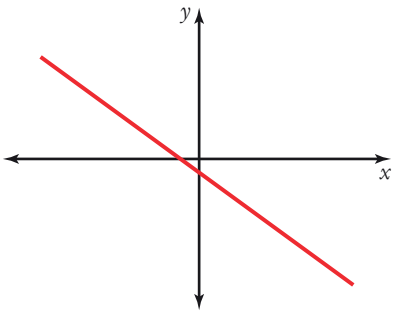
6 analysis questions
25 marks
Show your working where appropriate.

1 The graph of a function is shown on the right.
Use it to sketch a graph of the corresponding gradient function.



[4 marks]

2 The graph of a gradient function is shown on the right.
Use it to sketch a graph of the corresponding function.



[4 marks]

3 $f(x) = 2x^2 + x - 4$.

a Find the gradient of the tangent at $x = 1$.

b Find the equation of the tangent at $x = 1$.

[7 marks]

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4 The current I in amperes in a particular electrical circuit is given by $I = \frac{100}{R}$, where R is the resistance in ohms (Ω). Find the rate of change of I with respect to R when the resistance is 20 Ω.

[3 marks]

5 The volume of water flowing through a pipe is given by $V = t^2 + 3t$, where V is the volume in litres and t is the time in minutes. Find the rate of water flow after 5 minutes.

[3 marks]

6 For the graph of a function, $y = f(x)$, the rate of change of y values with respect to its x values is given by $\frac{dy}{dx} = 3x^2 - 2x + 1$. Given that the graph passes through the point $(-1, 3)$, find the equation of the function.

[4 marks]
Total marks: 65

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Answers

Part A [1 mark each]

1 A function is increasing when the derivative is positive.

$\therefore$ **C** **[1 mark]**

2 $y' = 3x^2 + 4x - 6$, so when $x = -2$, $y' = -2$

$\therefore y$ is decreasing.

$\therefore$ **D** **[1 mark]**

3 The derivative is the same as the gradient (slope). The function has a negative gradient when $x < 1$ and $x > 10$.

$\therefore$ **C** **[1 mark]**

4 $f'(x) = 8x - 3$ so $f'(3) = 21$.

$\therefore$ **B** **[1 mark]**

5 The derivative gives the rate of change. $f'(x) = 10x - 12$ so $f'(2) = 8$.

$\therefore$ **E** **[1 mark]**

6 $f(x) = 20x^4$, so $F(x) = \dfrac{20x^{4+1}}{5} + c = 4x^5 + c$.

$\therefore$ **A** **[1 mark]**

Part B

1 $f(x) = 2x^3 + 17x^2 - 12x + 8$

$f'(x) = 6x^2 + 34x - 12$ **[1 mark]**

a $f'(1) = 6 \times 1^2 + 34 \times 1 - 12 = 28$ **[1 mark]**

$f'(1) > 0 \therefore f(x)$ is increasing. **[1 mark]**

b $f'(-2) = 6 \times (-2)^2 + 34 \times (-2) - 12 = -56$ **[1 mark]**

$f'(-2) < 0$

$\therefore f(x)$ is decreasing. **[1 mark]**

c $f'\left(\frac{1}{3}\right) = 6 \times \left(\frac{1}{3}\right)^2 + 34 \times \frac{1}{3} - 12 = 0$ **[1 mark]**

$f'\left(\frac{1}{3}\right) = 0$

$\therefore f(x)$ is stationary. **[1 mark]**

[7 marks]

2 **a** $p'(x) = 8x^3 - 14x + 8$ **[1 mark]**

b $p'(-3) = 8 \times (-3)^3 - 14 \times (-3) + 8 = -166$ **[1 mark]**

[2 marks]

3 $g'(x) = -12x^2 + 4x + 5$ **[1 mark]**

a $5g'(x) = 5(-12x^2 + 4x + 5) = -60x^2 + 20x + 25$ **[1 mark]**

b $\frac{2}{3}g'(x) = \frac{2}{3}(-12x^2 + 4x + 5) = -8x^2 + 2\frac{2}{3}x + 3\frac{1}{3}$ **[1 mark]**

[3 marks]

4 $D[2f(x) - 7g(x)] = 2f'(x) - 7g'(x)$ **[1 mark]**

$= 2(4x^3 - 3x^2 - x) - 7(-3x^2 + 4x + 5)$ **[1 mark]**

$= 8x^3 - 6x^2 - 2x + 21x^2 - 28x - 35$ **[1 mark]**

$= 8x^3 + 15x^2 - 30x - 35$ **[1 mark]**

[4 marks]

5 **a** $p'(x) = 35x^4 - 24x^2 + 5$ **[1 mark]**

b $g'(u) = 15u^2 - 6 - 8u$ **[1 mark]**

[2 marks]

6 $y = (2x^3 + 5x)^2 = 4x^6 + 20x^4 + 25x^2$ **[1 mark]**

$y' = 24x^5 + 80x^3 + 50x$ **[1 mark]**

[2 marks]

7 $y = (3x - 2)^3 = 27x^3 - 54x^2 + 36x - 8$ **[1 mark]**

$y' = 81x^2 - 108x + 36$ **[1 mark]**

[2 marks]

8 **a** $f(1) = 4 \times 1^2 - 3 \times 1 + 5 = 6$ **[1 mark]**

$f(5) = 4 \times 5^2 - 3 \times 5 + 5 = 90$ **[1 mark]**

Average rate of change $= \frac{f(5) - f(1)}{5 - 1} = \frac{90 - 6}{4} = 21$ **[1 mark]**

b Instantaneous rate of change $= f'(x) = 8x - 3$ **[1 mark]**

i $f'(1) = 8 \times 1 - 3 = 5$ **[1 mark]**

ii $f'(5) = 8 \times 5 - 3 = 37$ **[1 mark]**

iii $f'(3) = 8 \times 3 - 3 = 21$ **[1 mark]**

c The average rates of change for the values checked is the average of the instantaneous rates for the endpoints of the intervals for the average rates of change. **[1 mark]**

[8 marks]

9 $F(x) = \frac{8x^{3+1}}{4} - \frac{4x^{1+1}}{2} - 7x + c = 2x^4 - 2x^2 - 7x + c$ **[1 mark]**

Two possible antiderivatives are $2x^4 - 2x^2 - 7x + 5$ and $2x^4 - 2x^2 - 7x - 6$. **[1 mark]**

[2 marks]

10 $y = 4x - \frac{6x^{2+1}}{3} - \frac{2x^{5+1}}{6} + c = 4x - 2x^3 - \frac{1}{3}x^6 + c$ **[1 mark + 1 mark for the + c]**

[2 marks]

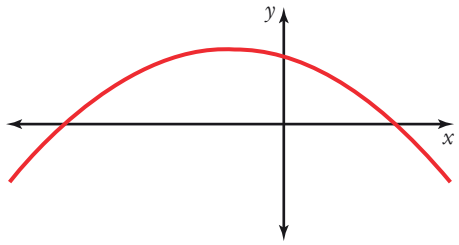
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Part C

1 The key features of the graph of the gradient function, $g(x)$, are found by examining the graph of the function, $f(x)$. These are:

- $f(x)$ is stationary on either side of the y-axis. This means that $g(x)$ will be zero at these x values.
  [1 mark for showing this feature on the graph]
- $f(x)$ has a positive slope between the stationary points. This means that $g(x)$ will be positive between these x values.
  [1 mark for showing this feature on the graph]
- Looking from left to right, $f(x)$ has a negative slope on the left of the first stationary point and on the right of the second stationary point. This means that $g(x)$ will be negative at these x values.
  [1 mark for showing this feature on the graph]
- The steepness of the graph of $g(x)$ is determined by the steepness of the tangent to the graph of $f(x)$.
  [1 mark for showing this feature on the graph]

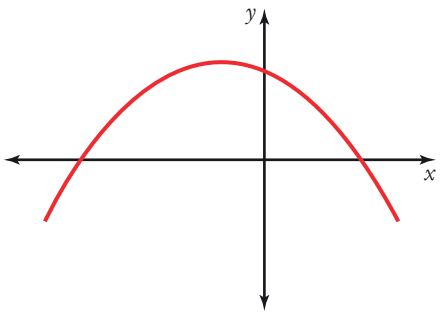
[4 marks]



2 The key features of the graph for the function, $f(x)$, are found by examining the graph of the gradient function, $g(x)$. These are:

- $g(x)$ is zero just to the left of the y-axis. This means that $f(x)$ will be stationary at this x value.
  [1 mark for showing this feature on the graph]
- $g(x)$ is positive to the left of the intercept with the x-axis. This means that $f(x)$ will have a positive gradient for these x values.
  [1 mark for showing this feature on the graph]
- $g(x)$ is negative to the right of the intercept with the x-axis. This means that $f(x)$ will have a negative gradient for these x values.
  [1 mark for showing this feature on the graph]
- The values of $g(x)$ indicate the steepness of the tangent to the graph of $f(x)$ − higher values of $g(x)$ indicate that the graph of $f(x)$ will be steeper than at lower values of $g(x)$.
  [1 mark for showing this feature on the graph]

[4 marks]



3 a $f'(x) = 4x + 1$ **[1 mark]**

$f'(1) = 4 \times 1 + 1 = 5$ **[1 mark]**

Gradient of tangent to $f(x) = 2x^2 + x - 4$ at $x = 1$ is 5. **[1 mark]**

b $f(1) = 2 \times 1^2 + 1 - 4 = -1$ **[1 mark]**

Tangent passes through $(1, -1)$ and has a gradient of 5.

Equation of tangent: $y - y_1 = m(x - x_1)$ **[1 mark]**

$[y - (-1)] = 5(x - 1)$ **[1 mark]**

$y = 5x - 6$ **[1 mark]**

[7 marks]

4 The rate of change of I with respect to $R = \frac{dI}{dR} = -\frac{100}{R^2}$ **[1 mark]**

When $R = 20\ \Omega$, $\frac{dI}{dR} = -\frac{100}{20^2} = -\frac{1}{4}$ **[1 mark]**

When $R = 20\ \Omega$, the current is decreasing at the rate of $\frac{1}{4}$ amperes per ohm. **[1 mark]**

[3 marks]

5 Rate of water flow: $V' = 2t + 3$ **[1 mark]**

Flow rate after 5 minutes $= 2 \times 5 + 3 = 13$ **[1 mark]**

After 5 minutes the water is flowing at 13 L/min. **[1 mark]**

[3 marks]

6 $f'(x) = 3x^2 - 2x + 1$

$f(x) = \frac{3x^{2+1}}{3} - \frac{2x^{1+1}}{2} + x + c = x^3 - x^2 + x + c$ **[1 mark]**

$f(-1) = 3 = x^3 - x^2 + x + c$ **[1 mark]**

$3 = (-1)^3 - (-1)^2 + -1 + c$

$3 = -1 - 1 - 1 + c$

$c = 6$ **[1 mark]**

$y = x^3 - x^2 + x + 6$ **[1 mark]**

[4 marks]

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